

Alison T. Cribb, PhD
1851 Research Fellow
School of Ocean and Earth Science, University of Southampton
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EDUCATION

- Ph.D. Geological Sciences, University of Southern California** 2018 - 2023
Los Angeles, California, USA
Doctoral dissertation title: “The geobiological role of bioturbating ecosystem engineers during key evolutionary intervals in Earth history”
Advisors: Prof. Frank A. Corsetti, Prof. David J. Bottjer
- B.A. Earth and Environmental Sciences, Vanderbilt University** 2014 - 2018
Conferred with Highest Honors
Nashville, Tennessee, USA
Honours thesis title: “Terminal Neoproterozoic bioturbation and implications for the extinction of the Ediacara biota”
Advisor: Dr. Simon A. F. Darroch

PROFESSIONAL APPOINTMENTS

- 1851 Research Fellow** 2023 - 2026
School of Ocean and Earth Science, University of Southampton
Funded by a Science and Engineering Research Fellowship from the Royal Commission for the Exhibition of 1851

AWARDS, HONORS, AND FELLOWSHIPS

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| Science and Engineering Research Fellowship | Royal Commission for the Exhibition of 1851 | 2023 - 2026 |
| PhD Achievement Award
<i>for academic excellence and success during doctoral studies, one of six awards across graduating PhDs</i> | University of Southern California | 2023 |
| Council Poster Prize
<i>for best poster among early career researchers</i> | Palaeontological Association | 2022 |
| Winifred Goldring Award, Honourable Mention | Association for Women Geoscientists; The Paleontological Society | 2022 |
| Gold Fellowship
<i>for research excellence among PhD candidates</i> | University of Southern California | 2021 |
| President's Prize
<i>for best talk among early career researchers</i> | Palaeontological Association | 2020 |

Cisco Systems Graduate Fellowship <i>awarded to a selected cohort of women pursuing doctorate degrees in science and engineering</i>	Women in Science and Engineering (WiSE) - University of Southern California	2020
Student Presentation Award, Honourable Mention	Geological Society of America, Geobiology and Geomicrobiology Division	2019
Degree conferred with Highest Honors	Vanderbilt University	2018

RESEARCH GRANTS

Science and Engineering Research Fellowship	Royal Commission for the Exhibition of 1851	2023 - 2026
Gold Fellowship	University of Southern California	2021
Graduate Student Research Grant	Geological Society of American	2019
Student Research Grant	Society for Sedimentary Geology (SEPM) Foundation	2019
Harry B. Whittington Award	Paleontological Society	2017
Eugene Vaughan Assistantship in Geology	Vanderbilt University	2017

PUBLICATIONS

*Supervised student manuscript

10. Darroch, S.A.F., Casey, M.M., **Cribb, A. T.**, Bates, A.E., Clapham, M.E., Contreras, D.L., Craffey, M., Duijnste, I.A.P., Gearty, W., Gotelli, N.J., Hamilton, M.J., Hayes, R.F., Hull, P.M., Ibarra, D.E., Lee, J., Looy, C.V., Lyson, T.R., Muddiman, B., Roopnarine, P.D., Shupinski, A.B., Smith, F.A., Stigall, A.L., Tome, C.P., Turk, K.A., Viallaseñor, A., Wagner, P.J., Wang, S., Lyons, S.K. 2025. Earth systems engineers and the lasting impact of organisms in deep time. *Trends in Ecology & Evolution* (doi: 10.1016/j.tree.2025.08.005)
9. Tarhan, L.G., Pippenger, K., **Cribb, A. T.**, Zill, M., Phelps, W., Savrda, C.E., Droser, M.L., Bottjer, D.J., Clapham, M.E. 2025. Tracking bioturbating through time: The evolution of the marine sedimentary mixed and transition layers. *Science Advances*, 11, eadu7719. (doi: 10.1126/sciadv.adu7719)
8. **Cribb, A. T.**, Darroch, S.A.F. 2024. How to engineer a habitable planet: the rise of marine ecosystem engineers through the Phanerozoic. *Palaeontology*, 67, e12726. (doi: 10.1111/pala.12726) (invited contribution)

7. **Cribb, A. T.**, Formoso, K., Woolley, H. C., Beech, J., Brophy, S., Byrne, P. J., Cassady, V. C., Godbold, A. L., Larina, E., Maxeiner, P., Wu, Y.H., Corsetti, F. A., and Bottjer, D. J. 2023. Contrasting terrestrial and marine ecospace dynamics after the end-Triassic mass extinction event. *Proceedings of the Royal Society B*, 290, 20232232. (doi: 10.1098/rspb.2023.2232)
6. **Cribb, A. T.**, van de Velde, S. J., Berelson, W. M., Bottjer, D. J., and Corsetti, F. A. 2023. Ediacaran-Cambrian bioturbation did not extensively oxygenate sediments in shallow marine ecosystems. *Geobiology*, 24, 435-453. (doi: 10.1111/gbi.12550)
5. Feng, X., Chen, Z., Benton, M. J., Su, C., Bottjer, D.J., **Cribb, A. T.**, Li, Z., Zhao, L., Zhu, G., Huang, Y., and Guo, Z. 2022. Resilience of infaunal ecosystems during the Early Triassic greenhouse Earth. *Science Advances*, 8, eabo0596 (doi: 10.1126/sciadv.abo0597).
4. Darroch, S. A. F., **Cribb, A. T.**, Buatois, L. A., Germs, G. J. B., Kenchington, C. G., Smith, E. F., Mocke, H., O'Neil, G. F., Schiffbauer, J., Maloney, K., Racicot, R. A., Turk, K. A., Gibson, B. M., Koester, B. E., Boag, T. H., Tweedt, S., and Laflamme, M. 2020. The trace fossil record of the Nama Group, Namibia: Exploring the Ediacaran roots of the Cambrian Explosion. *Earth-Science Reviews*, 103435. (doi: 10.1016/j.earscirev.2020.103435)
3. Maloney, K. M., Boag, T. H., Faccioli, A. J., Gibson, B. M., **Cribb, A. T.**, Koester, B. E., Kenchington, C. G., Racicot, R. A., Darroch, S. A. F., and Laflamme, M. Paleoenvironmental analysis of Ernietta-bearing Ediacaran deposits in southern Namibia. 2020. *Palaeogeography, Palaeoclimatology, Palaeoecology*, 556, 109884. (doi:10.1016/j.palaeo.2020.109884)
2. **Cribb, A. T.**, Bottjer, D. J. 2020. Complex marine bioturbation ecosystem engineering behaviors persisted in the wake of the end-Permian mass extinction. *Scientific Reports*, 10, 203. (doi:10.1038/s41598-019-56740-0)
1. **Cribb, A. T.**, Kenchington, C. G., Koester, B. E., Gibson, B. G., Boag, T. H., Racicot, R. A., Mocke, H., Laflamme, M., and Darroch, S. A. F. 2019. Increase in metazoan ecosystem engineering prior to the Ediacaran-Cambrian boundary in the Nama Group, Namibia. *Royal Society Open Science*, 6, 190548. (doi:10.1098/rsos.190548)

Published perspective, news, and opinion pieces

11. **Cribb, A. T.** & Darroch, S.A.F. 2024. Machine learning maps Earth's hidden bioturbation. *Current Biology*, 34, 630-632. (doi: 10.1016/j.cub.2024.05.038)

Software development and contributions

20. Stockey, R.G., **Cribb, A. T.**, Gruszka, P., Sartin, A., Monarrez, P. PaleoClimR. R package. (in development: github.com/richardstockey/PaleoClimR)

INVITED PRESENTATIONS AND SEMINARS

University of Bristol, School of Earth Sciences/School of Biological Sciences <i>Palaeontology seminar, Palaeobiology Research Group</i> Bristol, England, UK	2025
Cambridge University, Department of Earth Sciences <i>Department of Earth Sciences seminar</i> Cambridge, England, UK	2025
Geological Society of America Annual Meeting – Keynote speaker <i>Session: Phylogenetic and Computational Approaches in Paleobiology</i> Anaheim, California, USA	2024
University of Oxford, All Souls College and Department of Earth Sciences <i>“The Past and Future of Life on our Dynamic Planet” seminar series</i> All Souls College, University of Oxford, England, UK	2024
University of Portsmouth, School of the Environment, Geography, and Geosciences <i>School of the Environment, Geography, and Geoscience seminar</i> Portsmouth, England, UK	2022
University of Leeds, School of Earth and Environment <i>Earth Surface Science Institute (ESSI) seminar series</i> Leeds, England, UK	2022
Biosphere Evolution, Transitions and Resilience (BETR) – Keynote speaker Milton Keynes, England, UK	2022
6th Annual Nereis Park Conference – Plenary lecture <i>Session: Bioturbation-Ecosystem Relationships</i> The European Institute for Marine Sciences (IUEM), Lagonna-Daoulas, Brittany, France	2022
Goldschmidt Annual Meeting – Keynote speaker <i>Session: Life and the Environment in the Phanerozoic</i> Honolulu, Hawaii, USA	2022
NASA Jet Propulsion Laboratory <i>Origins and Habitability Lab seminar</i> Pasadena, California, USA	2020

TEACHING

Student mentorship

Katie Mills <i>Predicting coral reef latitudinal biodiversity gradients under various SSP scenarios using novel agent-based models</i>	MSci Marine Biology with Oceanography, University of Southampton	2025 - 2026
Bel Holmes <i>Coral extinction and recovery patterns at the Triassic-Jurassic boundary</i>	BSci Marine Biology, University of Southampton	2024 - 2025

Neha Ghosh

Terrestrial ecological responses to ancient and modern extinctions

MSci Natural Sciences (third-year), University of Southampton 2024 - 2025

Neha Ghosh

A Tale of Two Warming Events: Comparing ecological responses of modern and ancient climatic extremes

SMMI Undergraduate Summer Intern, University of Southampton 2024

Instructor

Earth and Ocean System

University of Southampton 2023, 2024

Palaeobiology

University of Southampton 2024

Marine Invertebrates

University of Southampton 2024

Co-evolution of Life and Earth

University of Southampton 2024

Teaching assistant

The History of Life on Earth: A View through the Museum

University of Southern California 2019, 2020, 2021

Climate Change

University of Southern California 2019

Crises of a Planet

University of Southern California 2018

Earth Systems Through Time

Vanderbilt University 2016, 2017

Pedagogical training

Practicum in Teaching the Liberal Arts

Focus on teaching in the geological and biological sciences

University of Southern California 2018

PROFESSIONAL SERVICE

Committees, appointments, and research networks

Secretary of the Earth System Science Group

Geological Society of London 2023-present

EDI Chair of the Earth System Science Group

Geological Society of London 2025-present

Ecological and Evolutionary Effects of Extinction and Ecosystem Engineers (E6) RCN

National Science Foundation-funded Research Coordination Network 2021-present

Graduate Student Representative

Geological Society of America – Geobiology and Geomicrobiology division 2020-2022

Dean's Student Advisory Board

Vanderbilt University 2016-2018

Academic meetings and seminars

International Fossil Coral and Reef Society, 4th Early Career Research Symposium

Organizing committee member

International Fossil Coral and Reef Society 2025

Life and Planet 2025

Organizing committee member

Geological Society of London 2025

Life and Planet 2024

Organizing committee member

Geological Society of London 2024

Symposium of Vertebrate Palaeontology and Comparative Anatomy

Organizing committee member

University of Southampton 2024

Life and Planet 2023

Organizing committee member

Geological Society of London 2023

Paleo/Environment Seminar

Co-organiser of weekly seminars for the academic year

University of Southern California 2021-2022

Southern California Geobiology Symposium

Deputy co-chair of organizing committee

University of Southern California (ran virtually) 2020-2021

Conference sessions

Ecosystem Engineering Through Earth History

Co-chaired with other graduate student members of the E6 RCN

Geological Society of America Denver, Colorado 2022

New Voices in Geobiology

Annual GSA GBGM Division Session

Geological Society of America Denver, Colorado 2022

New Advances in Geobiology

Annual GSA GBGM Division Session

Geological Society of America Denver, Colorado 2022

New Voices in Geobiology

Annual GSA GBGM Division Session

Geological Society of America Portland, Oregon 2021

New Advances in Geobiology

Annual GSA GBGM Division Session

Geological Society of America Portland, Oregon 2021

Ecosystem Engineering Through Earth History

Geological Society of America Virtual Meeting 2020

New Voices in Geobiology

Annual GSA GBGM Division Session

Geological Society of America Virtual Meeting 2020

New Advances in Geobiology

Annual GSA GBGM Division Session

Geological Society of America Virtual Meeting 2020

Phanerozoic Earth-Life Transitions	Geobiology Society Conference	2019
<i>Invited student session chair</i>	Banff, Canada	

Other service roles

President's Prize judge	The Palaeontological Association	2024
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Judge for PalAss President's Prize for best three student talks at annual meeting

Student presentation judge	Geological Society of America – Geomicrobiology and Geobiology Division	2023, 2024
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GSA Geobiology and Geomicrobiology Division student presentation awards at the GSA Annual meeting

Student presentation judge	The Paleontological Society	2023, 2024
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Paleontological Society student presentation awards at the GSA Annual Meeting

Reviewer

Current Biology; Geobiology; Geological Magazine; Journal of Open Source Science; Paleobiology; Palaeogeography, Palaeoclimatology, and Palaeoecology; Palaeontology; Proceedings of the National Academy of Sciences; Proceedings of the Royal Society B; Scientific Reports; Science Advances

Professional membership

Fellow of the Geological Society (FGS); Geological Society of London Earth System Science Group; the Palaeontological Association

PUBLIC OUTREACH AND ENGAGEMENT

Southampton Science and Engineering Festival (SOTSEF)	2025
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Activity leader: "Ocean Time Machine: Explore reef fossils and ancient climates"

Winchester Cathedral Whales	2025
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Design of evolution of whales and marine megafauna exhibit

Sidmouth Science Festival	2024
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*Volunteer with University of Southampton, School of Ocean and Earth Science
Developed and presented exhibit "Reefs Through Time"*

Southampton International Boat Show	2024
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*Volunteer
Exhibition of "Reefs Through Time" in University of Southampton tent*

Lyme Regis Fossil Festival	2024
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*Organizing team and volunteer with University of Southampton, School of Ocean and Earth Science
Developed and presented exhibit "Reefs Through Time"*

Southampton Science and Engineering Festival (SOTSEF)	2024
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*School of Ocean and Earth Science organizing team
Assisted in development of exhibit on coral reefs, mammoths, and ichthyosaurs*

Stem Bytes Seminar Speaker, Women in Science and Engineering (WiSE) chapter of the University of Southern California 2020

“How Early Worms Helped Engineer Earth’s Early Benthic Ecosystems”

Public talk aimed to increase recruitment of women into STEM majors at the University of Southern California.

On Point, National Public Radio (NPR) guest 2018

“It’s Senior Thesis Season” episode

Invited to discuss undergraduate honors thesis to a national audience on live radio

CONFERENCE ABSTRACTS

Abridged: First-author abstracts presented since 2020

[†]Invited presentation

[§]Award given for presentation

23. Cribb, A.T. 2025. What controlled the appearance of coral reef biodiversity hotspots throughout the Phanerozoic? *International Fossil and Coral Reef Society Early Career Symposium*. Online. Oral presentation.
22. Cribb, A.T. 2025. Responses and effects of marine ecosystem engineers during the end-Permian mass extinction. *Crossing the Paleontological-Ecological Gap & Conservation Paleobiology Symposium*, Zurich, Switzerland. Oral presentation.
21. Cribb, A.T. 2025. Ecological and climatic controls on coral reef biodiversity hotspots through the Phanerozoic. *Life and Planet*, London, England. Oral presentation. Highlight talk.
20. Cribb, A.T. 2024. Do ecosystem engineers facilitate recovery after mass extinctions? *Palaeontological Association Annual Meeting*. Erlangen, Germany. Poster presentation.
- [†]19. Cribb, A.T. 2024. Role of ecosystem engineers in Phanerozoic biodiversity and ecology. *Geological Society of America Annual Meeting*, Anaheim, CA. Oral presentation. Invited keynote for “Phylogenetic and Computational Approaches in Paleobiology.”
18. Cribb, A.T., Darroch, S.A.F., Gearty, W., Ezard, T.H.G. 2024. Phanerozoic marine biodiversity increased in presence of ecosystem engineers. *Life and Planet*, London, England. Poster presentation.
17. Cribb, A.T., Darroch, S.A.F., Ezard, T.H.G. 2024. Ecosystem engineers impact marine biodiversity through the Phanerozoic. *European Geosciences Union*, Vienna, Austria. Oral presentation (virtual).
16. Cribb, A.T., Stockey, R.G., Feng, X., Bottjer, D.J. 2023. Environmental drivers and biogeochemical consequences of benthic extinctions in the Early Triassic. *Geological Society of America Annual Meeting*, Pittsburgh, PA, USA. Oral presentation.
15. Cribb, A.T., Darroch, S.A.F., Ezard, T.H.G. 2023. How to engineer a habitable planet: New frameworks for understanding the rise and fall of marine ecosystem engineers through Earth history. *The Palaeontological Association Annual Meeting*, Cambridge, England. Oral presentation. Selected for the annual meeting symposium, “Ecosystem engineers through deep time”

14. Cribb, A.T., Stockey, R.G., Feng, X., Bottjer, D.J. 2023. Biogeochemical consequences of benthic extinctions during the end-Permian mass extinction. *Life and Planet*, London, England. Poster presentation.
13. Cribb, A.T., Formoso, K., Woolley, H.C., Beech, J., Brophy, S., Byrne, P.J., Cassady, V.C., Godbold, A.L., Larina, E., Maxeiner, P., Wu, Y.H., Corsetti, F.A., Bottjer, D.J. 2022. Decoupled terrestrial and marine ecological recovery after the end-Triassic mass extinction event. *Geological Society of America Annual Meeting*. Denver, CO. Hybrid oral presentation.
- [†]12. Cribb, A.T., van de Velde, S.J., Berelson, W.M., Corsetti, F.A., Bottjer, D.J. 2022. Investigating macrofauna-driven sediment oxygenation during the Ediacaran-Cambrian transition. *Biosphere, Evolutions, and Transitions (BETR) finale meeting*. Milton-Keynes, England. Invited oral presentation.
- [†]11. Cribb, A.T. Utilization of the trace fossil record to understand bioturbators' ecosystem engineering impact over the last 560 million years. *6th Nereis Park Conference and Thematic School*, Lagonna-Daoulas, France. Invited plenary speaker for session "Bioturbation-ecosystem relationships."
- [§]10. Cribb, A.T., Godbold, A.L., van de Velde, S.J., Koester, B.E., Celestian, A., Bottjer, D.J., Corsetti, F.A. 2022. Bioturbating ecosystem engineers and their sedimentary biogeochemical impact: A case study from the Lower Cambrian Deep Spring Formation. *The Palaeontological Association Annual Meeting*, Cork, Ireland. Poster presentation. Awarded Annual Meeting Council Poster Prize.
9. Cribb, A.T., Formoso, K., Woolley, C.H., Beech, J.D., Brophy, S., Byrne, P.J., Cassady, V.C., Godbold, A.L., Larina, E., Maxeiner, P., Wu, Y., Corsetti, F.A., Bottjer, D.J. 2022. Terrestrial and marine ecospace dynamics across the end-Triassic mass extinction event. *The Palaeontological Association Annual Meeting*, Cork, Ireland. Poster presentation.
- [†]8. Cribb, A.T., van de Velde, S.J., Berelson, W.M., Bottjer, D.J., Corsetti, F.A. 2022. Geobiological impacts of Earth's early complex bioturbators during the Ediacaran-Cambrian transition. *Goldschmidt Annual Meeting*, Hawaii, USA. Invited keynote speaker in session "Life and the Environment in the Phanerozoic."
7. Cribb, A.T., van de Velde, S.J., Berelson, W. M., Corsetti, F. A., Bottjer, D. J. 2022. Revealing the biogeochemical role of Ediacaran-Cambrian bioturbators during the Agronomic Revolution. *Southern California Geobiology Symposium*, Riverside, CA, USA. Oral presentation.
- [§]6. Cribb, A.T., van de Velde, S.J., Berelson, W.M., Corsetti, F.A., Bottjer, D.J. 2021. Did early bioturbators oxygenate shallow to deep sediment tiers in Ediacaran-Cambrian benthic ecosystems? *Geological Society of America Annual Meeting*, Portland, OR, USA. Oral presentation. Selected for The Paleontological Society's session Future Leaders in Paleontology.
5. Cribb, A.T., Bottjer, D.J. 2021. Biogeochemical consequences of a sedimentary mixed layer collapse in the wake of the end-Permian mass extinction. *Geological Society of America Annual Meeting*, Portland, OR, USA. Poster presentation.
4. Cribb, A.T., van de Velde, S.J., Berelson, W.M., Bottjer, D.J., Corsetti, F.A. 2021. Controls on the oxygen penetration depth in Ediacaran-Paleozoic benthic ecosystems: A reactive-transport modeling study. *Goldschmidt Annual Meeting*, online meeting. Online oral presentation.

- [§]3. Cribb, A.T., van de Velde, S.J., Darroch, S.A.F., Bottjer, D.J., Corsetti, F.A. 2020. Growing pains of the Agronomic Revolution: Early bioturbators stimulated sulfide production in shallow sediment tiers. *The Palaeontological Association Annual Meeting*, online meeting. Online oral presentation. Awarded the President's Prize.
2. Cribb, A.T., van de Velde, S.J., Darroch, S.A.F., Bottjer, D.J., Corsetti, F.A. 2020. Engineering benthic biogeochemistry: Reactive-transport modeling of biomixing and bioirrigation behaviors across the Ediacaran-Cambrian boundary. *Geological Society of America Meeting*, GSA Connects Online. Online oral presentation.
1. Cribb, A.T. & Bottjer, D.J. 2020. Complex benthic ecosystem engineering in the Early Triassic: The trace fossil record as a tool for understanding ecosystem function after mass extinction events. *Ocean Sciences Meeting*, San Diego, CA. Poster presentation.